Quotation No : Q26000119

Date : 19-Mar-2026

Scires(ciro Pharma)
Boduppal
HYDERABAD
TELANGANA

Kind Attention : Mr./Ms. Venu Pilli

Dear Sir,

We understand that in industry, reliability is not just a requirement,it is the foundation of success.
That is why Absstem Industrial PSA Generator is more than just a product; it is a commitment to excellence, efficiency, and uninterrupted performance.

Every component is designed with precision and innovation, ensuring that each molecule of oxygen and nitrogen is delivered with the utmost purity.

With fully automated operation, our system seamlessly integrates into your processes, reducing manual intervention and maximizing productivity.

Moreover, our industry-leading air ratio ensures that you achieve the highest efficiency, optimizing energy consumption while maintaining peak performance.
And because we believe in standing by our craftsmanship, we offer an unmatched-10 year warranty on sieves, giving you the peace of mind that your investment is protected for the long run.

We have attached the detailed quotation for your review.

Should you have any questions or require further customization to meet your specific needs, please do not hesitate to reach out.

We are here to provide not just a product but a solution that breathes life into your operations.

We look forward to the opportunity to serve you and build a lasting partnership.

Thanks and Regards,

For Absstem Industrial Solutions Pvt. Ltd.

Mohan Reddy N.V.
Regional Sales Manager
9289722207
www.absstem.com



nGEN POWERED BY ABSSTEM INDUSTRIAL

Industrial

PSA Nitrogen Generator

Unique Features of Absstem Nitrogen Generator nGEN UH

a. **Lowest Air to Gas ratio** in industry thereby guaranteeing **lowest power consumption**.
b. **10 years*** warranty on Molecular Sieves **(First In Industry)**.
c. Assured Performance for consistent Flow, Pressure and Purity of Nitrogen throughout Product Lifecycle.
d. IIOT ready machine having 100% operator free design for 24X7 operation.
e. Idle mode with auto start/stop features ensuring power savings during low/no demand.
f. Inbuilt dew point transmitter to monitor input compressed air quality as standard scope of supply.
g. Plug n Play machine with an ergonomic design suitable for accessibility.
h. All the Instruments are specially manufactured to make them suitable for Nitrogen gas.
i. All critical performance parameters are being monitored through HMI.
j. First in industry **Comprehensive Maintenance Contract (CMC)** service under Absstem Premium Shield program providing complete peace of mind to customer.

Ambient Design Parameters

a.	Temperature	0 to 45 Deg. Celsius
b.	Relative Humidity	up to 80 % (Max.)
c.	Altitude	up to 1000 m above Sea level

Absstem Industrial Solutions Pvt. Ltd.
Plot No. 591, Sector 8, Phase IV
IMT Manesar, Gurugram, Haryana - 122051
Contact - 1800 3010 3394 Email - indsales@absstem.com
CIN: U29256HR2022PTC103387

Product Description

a. Product Name	Ultra Purity PSA Nitrogen Gas Generator
b. PSA Module	**nGEN UH**
c. Purity Range	99.99% - 99.999%

General PSA Generation System Layout

A typical Industrial Nitrogen Generation system consists of three parts:

- Compressed Air System: Air Compressor, Pre-filter, Air Dryer, Dry Air Vessel
- PSA N_2 Generation System
- N_2 Storage Tank



Air Compressor	An Air compressor is used to compress the atmospheric air to a pressure required for the separation of **Nitrogen** from the air.
Refrigeration dryer	Compressed air then passes through a refrigerated dryer to remove moisture (which is harmful to sieves) from the air.
Air Tank	Dry air is collected in an air tank to remove the pulsations which serve the dynamic (changing) requirement of air by the PSA system.
Multistage Filtration system	Before air enters the PSA system, it is passed through multistage filtration removing moisture, particulates and oil content to the specified levels.
PSA system	Pressurized air is passed through PSA Towers filled with Molecular sieves to separate **Nitrogen** from Air.
Storage Tank	Industrial Grade **Nitrogen** produced from the PSA System is stored in this for continuous supply.
Oxygen Analyser	Nitrogen produced is monitored by an Oxygen analyser. In case, O_2 impurity goes above the specified level, it gives an alarm and switches the plant to vent mode.

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HD Touch Screen Display

(User-friendly touch screen interface for easy operation and maintenance)

Absstem industrial Nitrogen generators complies with all industry standards and regulation

- Alarm management system
- Process data display
- Consumables running hours' display
- Data logging via USB drive**
- Graphical representation**



Additional upgrades

- Digital flow meter with totalizer
- Nitrogen Pressure monitoring for High pressure booster driven plants
- Cylinder manifold and alarm management system
- Remote monitoring
- Nitrogen gas Dew-point monitoring

International Quality standards

A strict quality procedure, following the leading international pharmacopoeia and Industrial directives, is followed in the selection of all mechanical and electrical instruments, vessel fabrication and designing of Industrial Gas generators.

TÜV NORD ISO 9001 · TÜV NORD ISO 14001 · TÜV NORD ISO 13485 · RIR CERTIFICATION Partners in Growth

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Absstem Shield Premium

(Comprehensive Annual Maintenance Contract)



Any Industry is now heavily focused on increased OEE (Overall Equipment Efficiency) and reduced LCC (Life Cycle Cost) while deciding to invest in any equipment. At Absstem, we strive for an everlasting engagement with our customers by not selling them just a product but also provide a long term after-sales support.

We understand that to deliver a consistent performance, periodic health check and timely maintenance of the Nitrogen Generator is extremely important. With Absstem Shield Premium, you invest in complete peace of mind for all maintenance needs pertaining to N2 generators and associated integrated equipment like air compressor, Air dryer etc.

Few benefits of Absstem Shield Premium are listed as below: -

- Guaranteed assurance of plant performance.
- Genuine spare parts and consumables being used to provide longevity and durability.
- Regular visits by our technicians for health monitoring and carrying out the scheduled maintenance services.
- 24 x 7 On-Call service support to our customers for any basic trouble shooting
- Emergency visit to attend any unplanned breakdown to any location within 48 hours.
- No additional surprise charges as all breakdown spares and consumables covered.
- The contract can be signed at the time of equipment ordering itself and can be renewed on year on year basis.

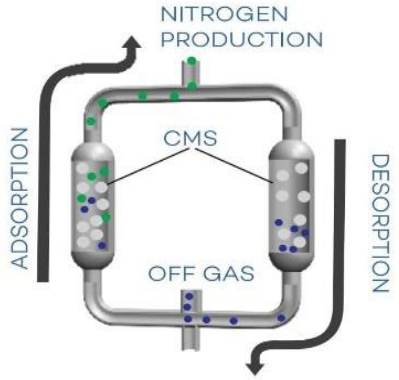
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nGENX POWERED BY ABSSTEM INDUSTRIAL

Industrial
PSA Nitrogen Generator



PSA Technology



PSA (PRESSURE SWING ADSORPTION)

Industrial Nitrogen Generator mainly consists of two adsorbers filled with carbon molecular sieves. Pressurized air is passed through one of these adsorbers, carbon having the property to attract oxygen, adsorbs oxygen from air and Nitrogen is allowed to pass through and move into the storage tank as the final product. As the adsorber approaches saturation, the air is allowed to pass through the other adsorber while this adsorber is being regenerated to ensure a continuous supply of Nitrogen. This changeover takes place through a specially designed microcontroller to ensure a hassle-free operation. The quality of Nitrogen produced is monitored by a Nitrogen analyser.

- Shipped ready for installation
- 5 minutes to run and never have to stop it!
- Easy installation
- Low operating expenses compared to gas cylinders/liquid Nitrogen
- Maintenance costs are very less
- Embedded Safety Interlocks
- Space-saving design
- Eliminates all hazards which are there with gas cylinder or liquid Nitrogen
- Longevity: at least 10 years

Description

a.	Product Name	Nitrogen Gas Generator

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b. Model Range nGENX
Technology PSA (Pressure Swing Adsorption)

Specifications

a.	Nitrogen Flow Rate	As per Performance Sheet
b.	Nitrogen Purity	As per Performance Sheet
c.	Nitrogen Delivery Pressure	6.5 Barg (Higher Pressure Options Available)
d.	Ambient conditions	
	Temperature	0 to 50 Deg. Celsius
	Relative Humidity	up to 80 % (Max.)
	Altitude	up to 1000 m above Sea level

Scheme Offered – (PSA Only)

This scheme will comprise of

- Air package with inlet air filtration system
  (Air compressor, Air dryer, Water Separator, Oil filters)
- Industrial Nitrogen generator
- Storage Tank
- Nitrogen Analyzer
- Alarm panel

Unique Features of Absstem

a. All the fabricated items are cleaned and degreased following our strict quality policy.
b. All the Instruments are specially manufactured to make them suitable for Nitrogen gas.
c. Guaranteed 10+ years of healthy life.
d. We provide the purest **Nitrogen** in Industry.
e. Data logging and Graphical Reports for all parameters available.
f. Test certificates of all instruments are available on demand.

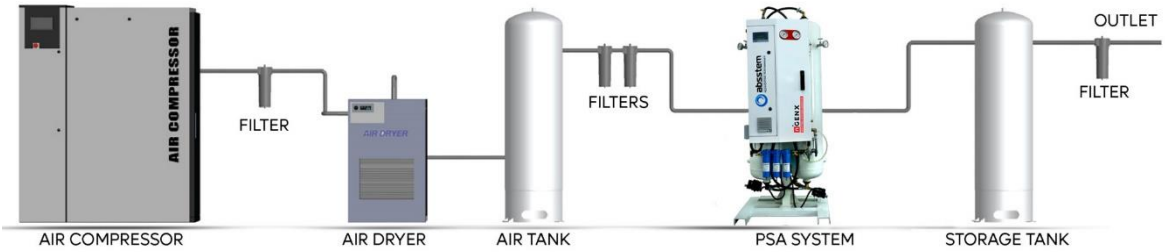
Onsite Gas Generation vs Liquid Tank/Cylinder Alternative

Non-stop availability: 24 hours a day, 7 days a week
Easy integration within existing installations
Your own independent supply of industrial gas
Significant economies for scale and lower operational costs: no rental charges, logistical expenses and bulk user evaporation losses
No safety hazards when handling high-pressure cylinders

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Equipment Description

The Industrial Nitrogen Generator consists of



Air Compressor	An Air compressor is used to compress the atmospheric air to a pressure required for the separation of **Nitrogen** from the air.
Refrigeration dryer	Compressed air then passes through a refrigerated dryer to remove moisture (which is harmful to sieves) from the air.
Air Tank	Dry air is collected in an air tank to remove the pulsations which serve the dynamic (changing) requirement of air by the PSA system.
Multistage Filtration system	Before air enters the PSA system, it is passed through multistage filtration removing moisture, particulates and oil content to the specified levels.
PSA system	Pressurized air is passed through PSA Towers filled with Molecular sieves to separate **Nitrogen** from Air.
Storage Tank	Industrial Grade **Nitrogen** produced from NGEN is stored in this tank to supply a continuous flow of **Nitrogen**.
Nitrogen Analyzer	Nitrogen produced is monitored by a **Nitrogen** analyser. In case, O2 purity goes below the specified level, it gives an alarm and switches it to vent mode.

EYE-SMART Control System

(User-friendly interface for easy operation and maintenance)

Absstem industrial Nitrogen generator complies with all industry standards and regulation

- Alarm management system
- Plug and play
- Fully automatic
- Process data display
- Consumables running hours display
- Data logging



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Additional upgrades

- Digital flow meter with totaliser
- High-pressure Nitrogen tank
- Cylinder manifold and alarm management system
- Remote monitoring- IoT Ready

Plant Operation

- Unit operates 24-hrs a day, 7-days a week reliably, automatically & safely

Automation

- The Plant is fully automatic and needs only electrical power for running the Air compressor
- The machine automatically enters into standby mode when the level in the Storage Tank reaches its set limit. This helps to increase the life of the Air compressor.

International Quality standards

A strict quality procedure, following the leading international pharmacopoeia and Industrial directives, is followed in the selection of all mechanical and electrical instruments, vessel fabrication and designing of Industrial Nitrogen generators.

Itemised Supply description

No.	Item description	Make	Qty
a	Air Compressor with Filters	ABSSTEM APPROVED	Client Scope / Quoted Separately
b	Refrigeration Dryer	ABSSTEM APPROVED	Client Scope / Quoted Separately
c	Air Tank	ABSSTEM	Client Scope / Quoted Separately
d	Set of Filters	ABSSTEM	1 no.
e	PSA unit (NGEN)	ABSSTEM	1 no.
f	Electrical Panel	ABSSTEM	1 no.
f	Digital Nitrogen Analyzer	ABSSTEM	1 no.
g	Storage Tank	ABSSTEM	1 no.
h	Outlet product gas Filter	ABSSTEM	1 no.

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Electrical Protection

The client should make proper arrangements for earthing/grounding systems for the Industrial **Nitrogen** Generator at their facility.

Enclosure Panel

- Ergonomically designed metal enclosure with powder-coated finish for longer life.
- High-strength IP-rated panel fittings from Germany for a robust enclosure.
- Fan filters IP-55 rated making the electrical enclosure's internal environment dust-free.
- Easy connectivity to any place of installation with req. port connections.
- High-strength base frame designed for higher load and easy transportation.
- Vessels are designed per ASME section VIII DIV I. Vessels are epoxy paint coated which is considered good for longevity and corrosion resistance.
- All electrical items are CE certified with operating voltage at 24 VDC

Annual Maintenance Contract (AMC)

Our in-house support Absstem shield is 24*7 open and we assure a timely service maintenance schedule. Per the schedule mentioned in the operational manual, we suggest the operator change the filters by following a simple protocol.

It is very easy to change or replace as our commissioning engineer plus our commissioning engineer and tech support office will always guide you about the activity.

Best ROI Guaranteed

ROI within 6 to 18 months when compared to conventional methods

Generator performance depends on the component selection. Our processes make this job easier as all equipment is selected based on more than 3 decades of research. NGEN is a fully automatic (operator-free operation) that ensures peace of mind.

We deliver what we commit!!

Thank You

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MINIMUM SPECIFICATIONS FOR FOOD GAS APPLICATIONS

Doc 126/18

Revision of Doc 126/11

EUROPEAN INDUSTRIAL GASES ASSOCIATION AISBL

AVENUE DES ARTS 3-5 • B – 1210 BRUSSELS
Tel: +32 2 217 70 98 • Fax: +32 2 219 85 14
E-mail: info@eiga.eu • Internet: www.eiga.eu

Appendix 1 - Summary of Current Gases Specifications in EU Legislation and JECFA

Component / Impurity	Standard	Carbon dioxide E 290	Nitrogen E 941	Oxygen E 948	Argon E 938	Helium E 939	Nitrous oxide E 942	n-/iso-Butane E 943 a/b	Propane E 944	Hydrogen E 949	Sulfur dioxide E 220
Assay (v/v)	EC	>99%	>99%	>99%	>99%	>99%	>99%	>96%/>94%	>95%	>99,9%	>99 %
	JECFA	>99%	>99%	>99%	>99%	>99%	>97%				>99,9 %
Odour	EC										
	JECFA			Free	free	free					
Moisture	EC		<0,05%	<0,05%	<0,05%	<0,05%	<0,05%	<50vppm	<50vppm	<50vppm	<0,05%
	JECFA	<52 vppm									<0,05%
CO_2	EC										
	JECFA			<300 vppm							
CO	EC	<10 vppm	<10 vppm				<30 vppm				
	JECFA	<10 vppm	<10 vppm	<10 vppm		<10 vppm	<10 vppm				
NO/NO_2	EC		<10 vppm				<10 vppm				
	JECFA						<5 vppm				
Total Hydrocarbon	EC		<100 vppm	<100 vppm	<100 vppm	<100 vppm		see *	see **		
	JECFA	<50vppm									
Residual Gases (O_2, N_2, H_2)	EC		< 1 % (O2)							<0.07%	
	JECFA				<1%						
Oil	EC	<5mg/kg									
	JECFA	<10 ppmw									
Acidity & Red. Subst.	EC	pass test									
	JECFA	pass test									
Halogens & H_2S	EC										
	JECFA						<5 vppm				
Other comp. & Heavy metals	EC										see***
	JECFA										see****

* Methane < 0,15%, other hydrocarbons < 5,1%. For E943a the maximum limit of hydrocarbons are:
C_2H_6 < 0,5%;
C_3H_8<1,5%;
i-C_4H_{10}<3%,
1.3-C_4H_6<0,1%

** Methane < 0,15%, other hydrocarbons are < 6,6%. For E943b the maximum limit of hydrocarbons are:
C_2H_6 < 0,5%;
C_3H_8<2,0%;
i-C_4H_{10}<4%,
1.3-C_4H_6<0,1%

*** EC Directive 2008/84 [2]: Non-volatile residue <0.001 %, Sulphur trioxide < 0,1 %, Selenium < 10 mg/kg, Arsenic <3mg/kg, Lead <5mg/kg, Mercury <1mg/kg, Heavy metals (as Pb) <10 mg/kg

**** JECFA (1998) [3]: Non-volatile residue <0,05%, Selenium <20 mg/kg, Lead <5mg/kg ,Other gases not present in air : No trace

Product Description

Product	Description	Make	Qty
nGen Nitrogen Generator Ultra Purity	nGEN007UH Nitrogen		1

Caption	Specification
Gas Generated	Nitrogen
Gas Pressure	upto 6.5 Barg
FLOW AT 99.99% PURITY	14 Nm3/Hr DRY AIR REQUIRED 40 CFM
FLOW AT 99.995% PURITY	13 Nm3/Hr DRY AIR REQUIRED 45 CFM
FLOW AT 99.999% PURITY	10 Nm3/Hr DRY AIR REQUIRED 47 CFM
COMPRESSED AIR PRESSURE REQUIRED	7.5 Barg
RECOMMENDED NITROGEN STORAGE	500 Liters
POWER SUPPLY	230 V/ 110 V 50/60 Hz Single Phase
RECOMMENDED AIR RECEIVER	500 Litres Part of Air System
INPUT AIR DEW POINT MONITOR	INCLUDED
CMS Warranty	10 Years*

Product	Description	Make	Qty
Gas Storage Vessel	AI-500L Storage Vessel		1

Caption	Specification
Water Volume	500 Litres
Gas Stored	Nitrogen/Oxygen
Design Standard	ASME Sec. VIII Div.1 Pressure Vessel Design
Working Pressure	8 Barg
Hydrotest Pressure	13.2 Barg
Accessories Included	Pressure Guage Safety Valve Manual Drain Additional Port for Other Instruments
Material of Construction	IS2062 Gr.B
Gas Stored Volume	3.25 M3
Corrosion Allowance	1.5 MM

Absstem Industrial Solutions Pvt. Ltd.
Ground Floor, Plot No. 591, Sector 8, Phase IV
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Product	Description	Make	Qty
nGenx Nitrogen Generator	nGenx 04 Nitrogen		1

Caption	Specification
Gas Generated	Nitrogen
Gas Pressure	upto 6.5 Barg(Higher options Available)
FLOW AT 99% PURITY	16 Nm3/Hr DRY AIR REQUIRED 21 CFM
FLOW AT 99.5% PURITY	14 Nm3/Hr DRY AIR REQUIRED 21 CFM
FLOW AT 99.9% PURITY	10 Nm3/Hr DRY AIR REQUIRED 19 CFM
FLOW AT 99.95% PURITY	9 Nm3/Hr DRY AIR REQUIRED 20 CFM
Module	PSA Plant
Air Compressor And Drier	Quoted Separately

Commercial Offer

S No	Item Code	Item Name	Description	HSN	Qty	Unit Price	GST(%)	Amount
1	nGEN007UH	nGEN007UH	nGen Nitrogen Generator Ultra Purity	840510	1	13,71,000.00	18.00	13,71,000.00
2	AIS0058ST	AI-500L	Gas Storage Vessel	731100	1	60,000.00	18.00	60,000.00
3	nGenx004	nGenx 04	nGenx Nitrogen Generator	840510	1	6,25,000.00	18.00	6,25,000.00
							Sub Total	20,56,000.00
							GST(%) @18.00	3,70,080.00
							Grand Total (INR)	24,26,080.00

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Terms And Conditions

1.	EXCLUSIONS & CLIENTS SCOPE	All erection / civil work, All Power Cables & Cabling out of battery limit of PSA Plant, Cable Trenches etc. Unloading / Shifting of Gas Plant at Clients site will be in Client Scope. Additional gas piping, cooling water (if required).
2.	SUPERVISION OF ERECTION & COMMISSIONING	There shall be no commissioning charges, however you shall arrange freeTo & Fro Air Ticket, lodging / boarding, and local conveyance for our Engineers during supervision of Installation of Gas Plant at site on free of cost basis. Our Engineers will be there for Supervision only. Any manpower / fitter required for installation work shall be provided by you.
3.	P&F	3% Extra.
4.	Freight	Extra at Actuals.
5.	Insurance	In your Scope.
6.	Payment Terms	50% Advance with your Prestegious Purchase Order and Balance against PI.
7.	Delivery	6-8 Weeks from the PO and Advance. Rest can be confirmed for Availability.
8.	Warranty	We hereby confirm that, our equipment shall have Suppliers warranty for a period of 12 months from the date of Commissioning or 18 months from date of Dispatch whichever is earlier.
9.	GST	As per the Commercial Table.
10.	Remarks	.

Hope you will find our offer in line with your requirement.

Should you require any further clarification/information, please feel free to contact us.

Thanking you, and assuring you of our best attention at all times.

Very Yours truly,

For Absstem Industrial Solutions Pvt. Ltd.

Mohan Reddy N.V.

Regional Sales Manager

9289722207

www.absstem.com